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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECTREPORT ON

SUPERMARKET CHECKOUT AND DISCOUNTS BILLING SYSTEM

Submitted by

HARSHINI.M

25BCS023

Under theGuidance of

Dr.Boopalan Sudaramoorthy

Technical Head

Technical Department, KG College of Arts and Science

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Semester: **II**

DECLARATION

I hereby declare that the project entitled "**SUPERMARKET CHECKOUT AND DISCOUNTS BILLING SYSTEM"** submitted in partial fulfillment of the requirements for the course **Java programming** is a **record of my original work** carried out under the guidance of **Dr. Boopalan Sundaramoorthy**, Technical Head, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

Ihave acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

HARSHINI.M

CERTIFICATE

This is to certify that the project entitled “SUPERMARKET CHECKOUT AND DISCOUNTS BILLING SYSTEM ” is a bona fide work carried out by Harshini.M,BSC CS 2025-2026, BSC.Computer science, KG College of Arts & Science, in partial fulfillment of the requirements for the course Java Programming language. This project has been completed under my supervision and guidance during the academic period April 2026.

Place: Coimbatore

Date:10/04/2026

Submitted for the Viva-Voce Examination held on 10/04/2026

Guide:
(Technical Head , KGCAS)

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Supermarket Checkout And Discount Billing System is a Java-based application developed using Swing GUI to simplify and automate the billing process in a supermarket. The system allows users to enter product details such as name, price, and discount percentage, and generates a bill in a structured format using a table layout. The application is designed using core Java concepts such as classes and objects to represent products and billing, encapsulation for secure data handling, inheritance and polymorphism to manage discounted products, and exception handling to handle invalid inputs. The system also uses file handling to store billing records for future reference. The graphical user interface provides a user-friendly environment where products can be added dynamically, and the total bill is calculated automatically. The use of a table structure enhances readability and mimics real-world supermarket billing systems. This project demonstrates the practical implementation of Java programming concepts in developing a real-time application, making it useful for small retail stores to manage billing efficiently and accurately

1. INTRODUCTION

In modern retail environments, manual billing systems are inefficient and prone to errors. To improve accuracy and speed, computerized billing systems are widely used in supermarkets. The Supermarket Checkout And Discount Billing System is developed to automate the billing process and make it faster, more reliable, and user-friendly. This project is implemented using Java programming language and Swing GUI, which provides an interactive interface for users to enter product details such as product name, price, and discount percentage. The system calculates the final price after applying the discount and generates a complete bill in a structured format using a table. The application uses core Java concepts such as classes and objects for designing the system, encapsulation for data security, inheritance and polymorphism for handling product variations, and exception handling to manage invalid inputs. File handling is also used to store billing details for future reference. Overall, this system reduces manual effort, minimizes calculation errors, and provides an efficient solution for supermarket billing operations.

1.1 Objective

The main objective of theSupermarketCheckout And Discount Billing System is to develop a simple and efficientbillingapplicationusing Core Java that automates the process of discount and bill generationinsupermarkets

The specific objectives of thisprojectare:

- To develop a user-friendlybillingsystemusingJava and Swing GUI.
- To automate theprocessofcalculatingproductprices with discounts.
- To reduce manualerrorsinbillingandimproveaccuracy.
- To implementcoreJavaconceptssuchasclasses, objects, inheritance, polymorphism, and encapsulation.
- Tohandleinvaliduserinputsusingexceptionhandling techniques.
- Tostorebillingdetailsusingfilehandlingforfuture reference.
- Todisplaythebillinastructuredtableformatsimilar to real supermarket systems.
- Toimproveefficiencyandspeedinthebillingprocess

1.2 Overview

The Supermarket Checkout And Discount Billing System is a Java-based desktop application designed to simplify and automate the billing process in supermarkets. It provides a graphical user interface using Java Swing, allowing users to easily enter product details such as product name, price, and discount percentage. The system processes the entered data and calculates
the final price after applying the discount. All products are displayed in a table format, which resembles a real supermarket billing system. Users can add multiple items, and the system automatically calculates the total amount for all products. The application is developed using core Java concepts such as object-oriented programming, including classes, objects, inheritance, polymorphism, and encapsulation. It also uses exception handling to manage invalid inputs and file handling to store billing records. Overall, the system provides a simple, efficient, and user-friendly solution for handling supermarket billing operations, making it suitable for small retail environments.

1.3 Features

Features – Supermarket Checkout and Discount Billing System

- User-Friendly Interface

  Provides an easy-to-use graphical interface using Java Swing for smooth interaction.

- Product Entry System

  Allows users to enter product name, price, and discount percentage during checkout.

- Table-Based Billing

  Displays all added items in a structured table format like a real supermarket bill.

- Automatic Discount Calculation

  Calculates the final price of each product after applying the discount automatically.

- Multiple Item Billing

  Supports adding multiple products and generates a combined total bill.

- AutomaticTotal Calculation

  Computes the total amount of all items instantly.

- ErrorHandling

  Handles invalid inputs (like letters instead of numbers) using exception handling.

- File Storage

  Saves billing details intoafile (bill.txt) for future reference.

- Fast and Efficient Processing

  Reduces manual work and speeds up the checkout

- process. Real-Time Output Display

  Shows billing details instantly on the screen as items are added.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently. The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

In traditional supermarket billing systems, manual calculations are time-consuming and prone to errors. Managing multiple products, applying discounts, and calculating the total bill manually can lead to inaccuracies and inefficiency. There is a need for an automated system that can handle product entry, apply discounts correctly, and generate accurate bills quickly. This project aims to solve these problems by developing a computerized billing system.

2.2 Project Overview

The Supermarket Checkout and Discount Billing System is a Java-based application developed using Swing GUI. It allows users to enter product details such as name, price, and discount percentage during checkout. The system calculates the final price after applying discounts and displays all items in a table format similar to real supermarket billing systems. It also calculates the total bill automatically and stores the billing details using file handling. The system is designed using object-oriented programming concepts for better structure and efficiency. This project is implemented using Core Java programming concepts and can be developed as either a console-based program or a Java Swing GUI application.

2.3 Scopes and Limitations Scope

The scope of the Supermarket Checkout And Discount Billing System includes the following:

- Can be used in small supermarkets and retail shops
- Supports multiple product entries
- Provides automatic discountcalculation
- Generates a structured billintable format
- Storesbillingrecords forfuture reference
- Easytousewith a graphicaluser interface

• Food courts
• Small food businesses

Limitations

- Limited to basic billing functions (no barcode scanning)
- Does not support database storage (uses only file storage)
- No multi-user or login system
- Suitable only for small-scale applications
- No online payment integration

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier. The Supermarket Checkout and Discount Billing System is designed as a GUI-based application where users can enter product details, apply discounts, and generate a final bill automatically. The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the Supermarket Checkout and Discount Billing System are explained below:

1. Product Entry Module

This module is responsible for entering product details such as product name, price, and discount percentage.

Example Input:

Product Name: Rice

Price: ₹100

Discount: 10%

The user enters these detailsthrough the graphical interface.

2. Item Addition Module

This module allows the userto add products to the billing table.

Functions of this module include:

- Accepting product details
- Calculating final price after discount
- Adding product details to the table
- Allowing multiple items to be added The system supports adding multiple products before generating the final bill.

3. Bill Calculation Module

This moduleperformsall thebilling calculations.

The systemcalculates:

- Finalpriceofeachproductafter discount

● Total amount of all products

The calculation is performed automatically using Java logic.

Example Calculation:

Final Price = Price – (Price × Discount / 100)

Total Bill = Sum of allfinal prices

4. Bill Display Module

Thismoduledisplaysthebill in a structured table format.

The bill includes:

● Product Name

● Price

● Discount (%)

● Final Price

Italsodisplaysthetotalbill amount at the bottom.

Example Output:

------SUPERMARKET BILL ------

Item Rice	Price 100		Discount 10%	Final Price 90
Milk	50	5%	47.5	

Total: 137.5

5. File Storage Module

This module saves the billing details into a file (bill.txt).

Ithelps in:

●Storing bill records

●Viewing previous transactions

6.Exit Module

This module allows the user to exit the application safely after completing the billing process. When the user closes the window, the system terminates the program.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Supermarket Checkout and Discount Billing System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of supermarket billing operations.

The main problems identified in manual billing systems are:

- Manual calculations may lead to billing errors.
- Preparing bills manually takes more time.
- Difficult to handle multiple customer orders efficiently.
- Lack of automated records for transactions.

Based on these problems, the system requirements were defined to develop an automated billing application that simplifies restaurant billing operations.

4.2. System Design

In this stage,theoverallstructure ofthe system is designed before coding begins.

The systemisdividedinto modulessuch as:

•Product Entry Module

● Item Addition Module

● Bill Calculation Module

● Bill Display Module

● File Storage Module

● Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language**.

The following steps were followed during development:

• Creation of classes and methods
• Implementation of menu-driven logic
• Writing code for bill calculation
• Handling user input
• Displaying the final bill

Core Java concepts such as **classes,objects,loops,conditionalstatements, and methods** were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Enter product details Add multiple items	Product added successfully All items displayed in table
Generate total Invalid input	Correct total

l	h
d	a
i	n
s	d
p	l

l e

a d

y p

e r

d o

E p

r e

r r

o l

r y

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates the final bill based on the selected food items and quantities.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the Supermarket Checkout and Discount Billing System:Start the program.

Start the program

Enter product name, price, and discount

Calculate final price using discount formula

Add product details to the table.

Repeat steps 2–4 for multiple items

Calculate total bill amount 7.

Display the bill in table format

Save the bill to file

9. End the program Display the final bill to the user.

10. End the program.

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start
↓
Enter Product Details

↓
Add to Table
↓
Enter Quantity
↓
Calculate Total
↓
Display Bill
↓
End
Students can draw using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

Include important parts of the Java code such as:

Include important parts of the Java code such as:

Main Class

```
public class SupermarketGUI {
public static void main(String[] args) {
new SupermarketGUI();
    }
}
```

Product Entry Logic

```
String name = nameField.getText();
double price = Double.parseDouble(priceField.getText());
double discount = Double.parseDouble(discountField.getText());
```

Billing Calculation Logic

```
double finalPrice = price - (price * discount / 100);
```

Total Calculation

```
double total = 0;
for (int i = 0; i < model.getRowCount(); i++) {
  total += (double) model.getValueAt(i, 3);
}
```

File Handling

```
FileWriter fw = new FileWriter("bill.txt");
fw.write("Total: " + total);
fw.close();
```

6. INPUT AND OUTPUT SCREENS

Figure 1: Main Screen

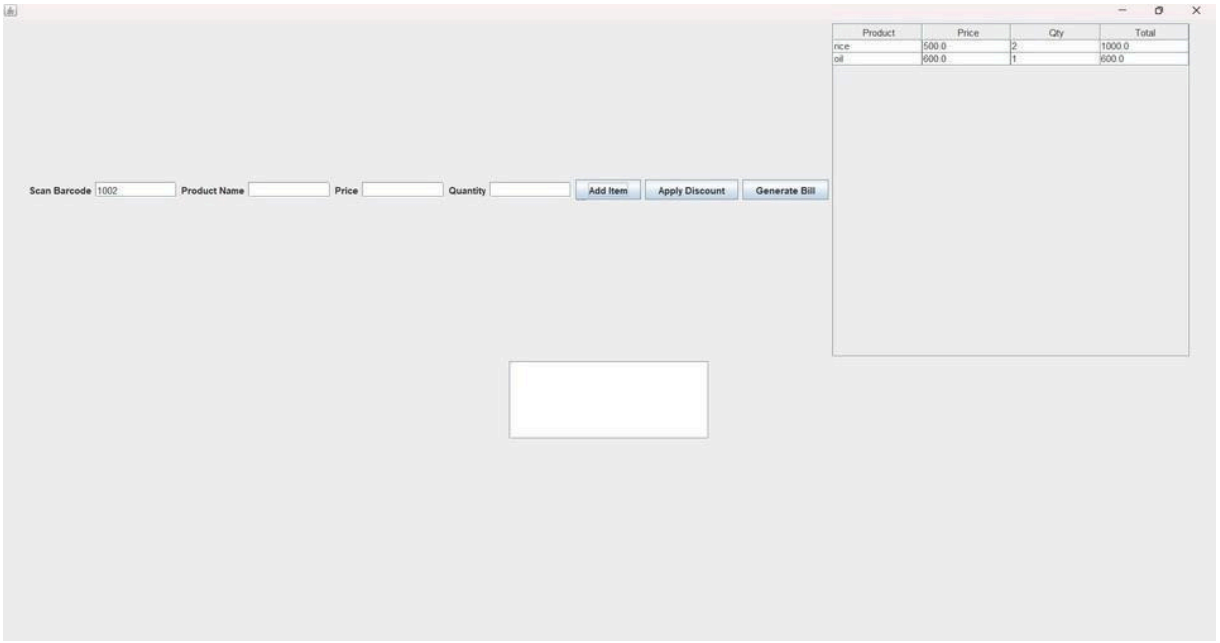


Figure 2: Adding Discount

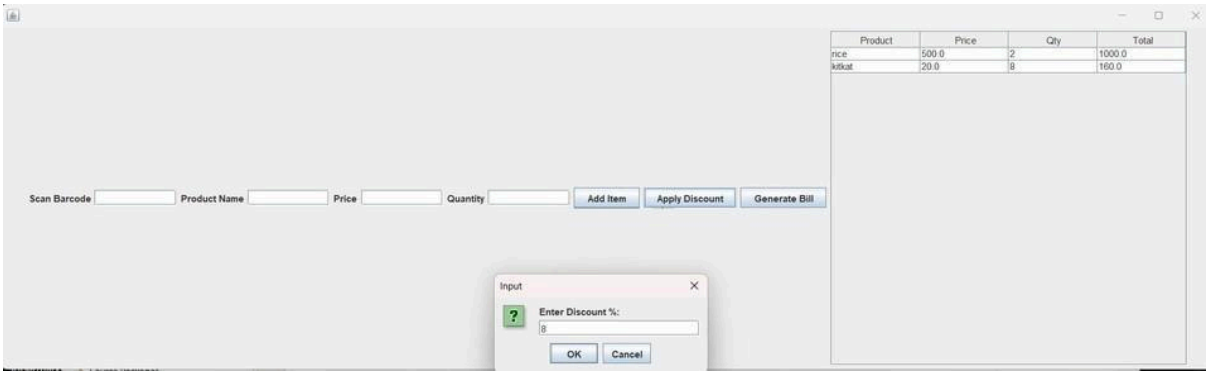
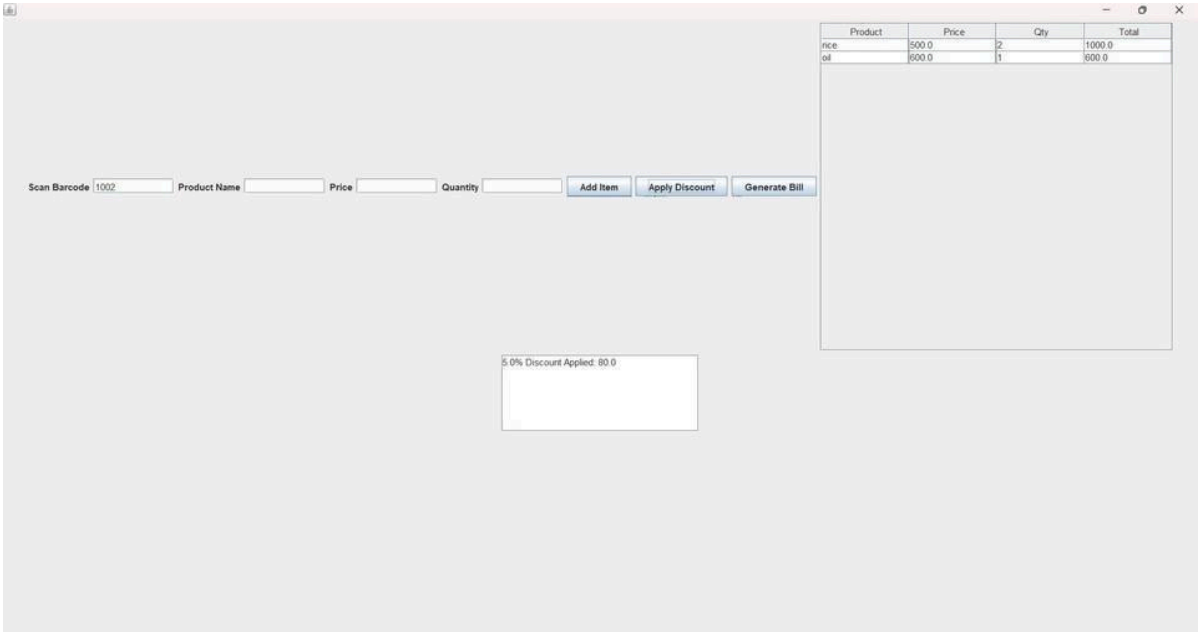


Figure 3: Discount added



7. DISCUSSION & CONCLUSION

7.1 Discussion

The Supermarket Checkout and Discount Billing System was developed to automate the checkout and billing process in supermarkets. The system allows users to enter product details such as name, price, and discount, and generates the bill automatically in a structured table format. During the development process, Core Java programming concepts such as classes, objects, inheritance, polymorphism, loops, conditional statements, and exception handling were applied. The system successfully performs discount calculations and generates accurate billing details. The system was tested with different inputs to ensure correct functionality. The results show that the system reduces manual effort, minimizes errors, and improves billing efficiency.

7.2 Conclusion

The Supermarket Checkout and Discount Billing System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world application. The system automates the billing process, applies discounts accurately, and generates bills quickly. It improves efficiency and reduces manual errors in supermarket checkout operations. This project helps in understanding how Java can be used to develop user-friendly and efficient software systems. The system can be further enhanced by adding features such as database integration, barcode scanning, and online payment options.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

•https://www.oracle.com/java
•https://www.geeksforgeeks.org
•https://www.javatpoint.com